EXP 5

Exploratory Data Analysis & Data Inspecting

AIM:

To inspect a car dataset using functions such as head(), tail(), shape, columns, dtypes, info() and describe(), to filter the data using conditions, and to find descriptive statistics (mean, median, mode) and plot simple charts.

ALGORITHM:

1. Import pandas, numpy and matplotlib, and load data.csv into a DataFrame.
2. Display the first five rows using head() and the last five rows using tail().
3. Display the number of rows and columns using shape.
4. Display the column names and the data type of each column using columns and dtypes.
5. Display the summary of the dataset using info() and the statistics of numerical columns using describe().
6. Filter the cars that have Highway MPG greater than 30.
7. Filter the BMW cars that have Engine HP greater than 200.
8. Filter the cars that have an Automatic transmission.
9. Select only the Make and MSRP columns.
10. Find the descriptive statistics of Engine HP, MSRP, city mpg and highway MPG.
11. Plot a bar chart of the average MSRP by vehicle size.
12. Plot a histogram and a box plot of Engine HP.
13. Calculate the mean, median and mode of Engine HP and show them as lines on the histogram.

CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# ------------------------------------------------------------------------------------
# Load Dataset
# ------------------------------------------------------------------------------------

df = pd.read_csv("data.csv")

# ------------------------------------------------------------------------------------
# 1. HEAD
# ------------------------------------------------------------------------------------

print("FIRST 5 ROWS:")
print(df.head())
```

```
# ------------------------------------------------------------------------------------
# 2. TAIL
# ------------------------------------------------------------------------------------

print("\nLAST 5 ROWS:")
```

```
print(df.tail())
```

```
# ------------------------------------------------------------------------------
# 3. SHAPE
# ------------------------------------------------------------------------------

print("\nSHAPE OF DATASET:")
print(df.shape)
```

```
# ------------------------------------------------------------------------------
# 4. COLUMNS.TOLIST()
# ------------------------------------------------------------------------------

print("\nCOLUMN NAMES:")
print(df.columns)
```

```
# ------------------------------------------------------------------------------
# 5. DTYPES
# ------------------------------------------------------------------------------

print("\nDATA TYPES:")
print(df.dtypes)
```

```
# ------------------------------------------------------------------------------
# 6. INFO
# ------------------------------------------------------------------------------

print("\nDATASET INFORMATION:")
df.info()
```

```
# ------------------------------------------------------------------------------
# 7. DESCRIBE
# ------------------------------------------------------------------------------

print("\nDESCRIPTIVE STATISTICS:")
print(df.describe())
```

```
# ------------------------------------------------------------------------------
# 8. Cars with Highway MPG > 30
# ------------------------------------------------------------------------------

highway_mpg = df[df['highway MPG'] > 30]

print("\nCARS WITH HIGHWAY MPG > 30:")
print(highway_mpg)
```

```
# ------------------------------------------------------------------------------
# 9. BMW Cars with Engine HP > 200
# ------------------------------------------------------------------------------

bmw_hp = df[
    (df['Make'] == 'BMW') &
    (df['Engine HP'] > 200)
]

print("\nBMW CARS WITH ENGINE HP > 200:")
print(bmw_hp)
```

```
# ------------------------------------------------------------------------------
# 10. Cars with Automatic Transmission
# ------------------------------------------------------------------------------

automatic_cars = df[
    df['Transmission Type'] == 'AUTOMATIC'
]

print("\nCARS WITH AUTOMATIC TRANSMISSION:")
print(automatic_cars)
```

```
# ------------------------------------------------------------------------------
# 11. Select Make and MSRP columns
# ------------------------------------------------------------------------------

selected_columns = df[['Make', 'MSRP']]

print("\nMAKE AND MSRP:")
print(selected_columns)
```

```
# ------------------------------------------------------------------------------
# 12. Descriptive Statistics
# ------------------------------------------------------------------------------

numeric_columns = [
    'Engine HP',
    'MSRP',
    'city mpg',
    'highway MPG'
]

print("\nDESCRIPTIVE STATISTICS OF SELECTED COLUMNS:")
print(df[numeric_columns].describe())
```

```
# ------------------------------------------------------------------------------
# 13. Bar Chart:
#     Average MSRP by Vehicle Size
# ------------------------------------------------------------------------------

average_msrp = df.groupby('Vehicle Size')['MSRP'].mean()

plt.figure(figsize=(8, 5))
plt.bar(
    average_msrp.index,
    average_msrp.values
)

plt.xlabel('Vehicle Size')
plt.ylabel('Average MSRP')
plt.title('Average MSRP by Vehicle Size')
plt.xticks(rotation=30)
plt.show()
```

```
# ------------------------------------------------------------------------------
# 14. Histogram:
#     Distribution of Engine HP
# ------------------------------------------------------------------------------
```

```
hp_data = df['Engine HP'].dropna()

plt.figure(figsize=(8, 5))
plt.hist(
    hp_data,
    bins=20,
    edgecolor='black'
)

plt.xlabel('Engine HP')
plt.ylabel('Frequency')
plt.title('Distribution of Engine HP')
plt.show()
```

```
plt.figure(figsize=(8,6))
plt.boxplot(hp_data, patch_artist=True,
            boxprops=dict(facecolor='lightgreen'))
plt.title('Boxplot of Engine HP')
plt.ylabel('Engine HP')
plt.grid(axis='y', alpha=0.6)
plt.show()
```

```
mean_hp = hp_data.mean()
median_hp = hp_data.median()
mode_hp = hp_data.mode()[0]

plt.figure(figsize=(9, 5))

plt.hist(hp_data, bins=20,color='lightgray', edgecolor='black')

plt.axvline(mean_hp, color='red', linewidth=2.5,
            label=f'Mean = {mean_hp:.2f}')

plt.axvline(median_hp, color='blue', linewidth=2.5,
            label=f'Median = {median_hp:.2f}')

plt.axvline(mode_hp, color='green', linewidth=2.5,
            label=f'Mode = {mode_hp:.2f}')

plt.xlabel('Engine HP')
plt.ylabel('Frequency')
plt.title('Engine HP Distribution with Mean, Median and Mode')
plt.legend()
plt.grid(axis='y', alpha=0.3)

plt.show()
```

OUTPUT:

Output of: import pandas as pd

```
FIRST 5 ROWS:
  Make       Model  Year             Engine Fuel Type  Engine HP  \
0  BMW  1 Series M  2011  premium unleaded (required)      335.0
1  BMW    1 Series  2011  premium unleaded (required)      300.0
2  BMW    1 Series  2011  premium unleaded (required)      300.0
```

```
3  BMW    1 Series  2011  premium unleaded (required)       230.0
4  BMW    1 Series  2011  premium unleaded (required)       230.0

   Engine Cylinders  Transmission Type      Driven_Wheels  Number of Doors  \
0               6.0             MANUAL  rear wheel drive              2.0
1               6.0             MANUAL  rear wheel drive              2.0
2               6.0             MANUAL  rear wheel drive              2.0
3               6.0             MANUAL  rear wheel drive              2.0
4               6.0             MANUAL  rear wheel drive              2.0

                        Market Category Vehicle Size Vehicle Style  \
0  Factory Tuner,Luxury,High-Performance       Compact         Coupe
1                     Luxury,Performance       Compact   Convertible
2                Luxury,High-Performance       Compact         Coupe
3                     Luxury,Performance       Compact         Coupe
4                                Luxury       Compact   Convertible

   highway MPG  city mpg  Popularity    MSRP
0           26         19         3916  46135
1           28         19         3916  40650
2           28         20         3916  36350
3           28         18         3916  29450
4           28         18         3916  34500
```

Output of: print("\nLAST 5 ROWS:")

```
LAST 5 ROWS:
          Make    Model  Year                 Engine Fuel Type  Engine HP  \
11909    Acura      ZDX  2012     premium unleaded (required)       300.0
11910    Acura      ZDX  2012     premium unleaded (required)       300.0
11911    Acura      ZDX  2012     premium unleaded (required)       300.0
11912    Acura      ZDX  2013  premium unleaded (recommended)       300.0
11913  Lincoln   Zephyr  2006                regular unleaded       221.0

       Engine Cylinders  Transmission Type       Driven_Wheels  Number of Doors  \
11909               6.0          AUTOMATIC    all wheel drive              4.0
11910               6.0          AUTOMATIC    all wheel drive              4.0
11911               6.0          AUTOMATIC    all wheel drive              4.0
11912               6.0          AUTOMATIC    all wheel drive              4.0
11913               6.0          AUTOMATIC  front wheel drive              4.0

                  Market Category Vehicle Size  Vehicle Style  highway MPG  \
11909  Crossover,Hatchback,Luxury       Midsize  4dr Hatchback           23
11910  Crossover,Hatchback,Luxury       Midsize  4dr Hatchback           23
11911  Crossover,Hatchback,Luxury       Midsize  4dr Hatchback           23
11912  Crossover,Hatchback,Luxury       Midsize  4dr Hatchback           23
11913                      Luxury       Midsize          Sedan           26

       city mpg  Popularity   MSRP
11909        16         204  46120
11910        16         204  56670
11911        16         204  50620
11912        16         204  50920
11913        17          61  28995
```

Output of: print("\nSHAPE OF DATASET:")

```
SHAPE OF DATASET:
(11914, 16)
```

Output of: print("\nCOLUMN NAMES:")

```
COLUMN NAMES:
Index(['Make', 'Model', 'Year', 'Engine Fuel Type', 'Engine HP',
       'Engine Cylinders', 'Transmission Type', 'Driven_Wheels',
       'Number of Doors', 'Market Category', 'Vehicle Size', 'Vehicle Style',
       'highway MPG', 'city mpg', 'Popularity', 'MSRP'],
      dtype='str')
```

Output of: print("\nDATA TYPES:")

```
DATA TYPES:
Make                        str
Model                       str
Year                      int64
Engine   Fuel   Type        str
Engine HP               float64
Engine   Cylinders      float64
Transmission Type           str
Driven_Wheels               str
Number of Doors         float64
Market Category             str
Vehicle Size                str
Vehicle Style               str
highway MPG               int64
city mpg                  int64
Popularity                int64
MSRP                      int64
dtype: object
```

Output of: print("\nDATASET INFORMATION:")

```
DATASET INFORMATION:
<class 'pandas.DataFrame'>
RangeIndex: 11914 entries, 0 to 11913
Data columns (total 16 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Make                11914 non-null  str
 1   Model               11914 non-null  str
 2   Year                11914 non-null  int64
 3   Engine Fuel Type    11911 non-null  str
 4   Engine HP           11845 non-null  float64
 5   Engine Cylinders    11884 non-null  float64
 6   Transmission Type   11914 non-null  str
 7   Driven_Wheels       11914 non-null  str
 8   Number of Doors     11908 non-null  float64
 9   Market Category     8172 non-null   str
 10  Vehicle Size        11914 non-null  str
 11  Vehicle Style       11914 non-null  str
 12  highway MPG         11914 non-null  int64
 13  city mpg            11914 non-null  int64
 14  Popularity          11914 non-null  int64
```

```
 15  MSRP               11914 non-null  int64
dtypes: float64(3), int64(5), str(8)
memory usage: 2.4 MB
```

Output of: print("\nDESCRIPTIVE STATISTICS:")

```
DESCRIPTIVE STATISTICS:
                Year    Engine HP  Engine Cylinders   Number of Doors  \
count   11914.000000  11845.00000      11884.000000     11908.000000
mean     2010.384338    249.38607          5.628829         3.436093
std         7.579740    109.19187          1.780559         0.881315
min      1990.000000     55.00000          0.000000         2.000000
25%      2007.000000    170.00000          4.000000         2.000000
50%      2015.000000    227.00000          6.000000         4.000000
75%      2016.000000    300.00000          6.000000         4.000000
max      2017.000000   1001.00000         16.000000         4.000000

         highway MPG      city mpg    Popularity          MSRP
count   11914.000000  11914.000000  11914.000000  1.191400e+04
mean       26.637485     19.733255   1554.911197  4.059474e+04
std         8.863001      8.987798   1441.855347  6.010910e+04
min        12.000000      7.000000      2.000000  2.000000e+03
25%        22.000000     16.000000    549.000000  2.100000e+04
50%        26.000000     18.000000   1385.000000  2.999500e+04
75%        30.000000     22.000000   2009.000000  4.223125e+04
max       354.000000    137.000000   5657.000000  2.065902e+06
```

Output of: highway_mpg = df[df['highway MPG'] > 30]

```
CARS WITH HIGHWAY MPG > 30:
          Make        Model  Year                Engine Fuel Type  Engine HP  \
32        FIAT  124 Spider  2017  premium unleaded (recommended)      160.0
33        FIAT  124 Spider  2017  premium unleaded (recommended)      160.0
34        FIAT  124 Spider  2017  premium unleaded (recommended)      160.0
41         BMW     2 Series  2016     premium unleaded (required)      240.0
42         BMW     2 Series  2016     premium unleaded (required)      240.0
...        ...          ...   ...                             ...        ...
11830   Toyota        Yaris  2017               regular unleaded      106.0
11831   Toyota        Yaris  2017               regular unleaded      106.0
11894      BMW           Z4  2014     premium unleaded (required)      240.0
11897      BMW           Z4  2015     premium unleaded (required)      240.0
11901      BMW           Z4  2016     premium unleaded (required)      240.0

       Engine Cylinders Transmission Type       Driven_Wheels  Number of Doors  \
32                  4.0            MANUAL   rear wheel drive              2.0
33                  4.0            MANUAL   rear wheel drive              2.0
34                  4.0            MANUAL   rear wheel drive              2.0
41                  4.0         AUTOMATIC   rear wheel drive              2.0
42                  4.0         AUTOMATIC   rear wheel drive              2.0
...                 ...               ...                ...              ...
11830               4.0         AUTOMATIC  front wheel drive              4.0
11831               4.0            MANUAL  front wheel drive              4.0
11894               4.0            MANUAL   rear wheel drive              2.0
11897               4.0            MANUAL   rear wheel drive              2.0
11901               4.0            MANUAL   rear wheel drive              2.0

          Market Category Vehicle Size  Vehicle Style  highway MPG  city mpg  \
```

```
32            Performance      Compact    Convertible           35        26
33            Performance      Compact    Convertible           35        26
34            Performance      Compact    Convertible           35        26
41     Luxury,Performance      Compact          Coupe           35        23
42                Luxury      Compact    Convertible           34        23
...                  ...          ...            ...          ...       ...
11830           Hatchback      Compact  4dr Hatchback           35        30
11831           Hatchback      Compact  4dr Hatchback           36        30
11894  Luxury,Performance      Compact    Convertible           34        22
11897  Luxury,Performance      Compact    Convertible           34        22
11901  Luxury,Performance      Compact    Convertible           34        22

       Popularity   MSRP
32             819  27495
33             819  24995
34             819  28195
41            3916  32850
42            3916  38650
...            ...    ...
11830         2031  18000
11831         2031  17200
11894         3916  48950
11897         3916  48950
11901         3916  49700

[2905 rows x 16 columns]
```

Output of: bmw_hp = df[

```
BMW CARS WITH ENGINE HP > 200:
      Make        Model  Year             Engine Fuel Type   Engine HP  \
0      BMW  1 Series M  2011  premium unleaded (required)       335.0
1      BMW    1 Series  2011  premium unleaded (required)       300.0
2      BMW    1 Series  2011  premium unleaded (required)       300.0
3      BMW    1 Series  2011  premium unleaded (required)       230.0
4      BMW    1 Series  2011  premium unleaded (required)       230.0
...    ...          ...   ...                           ...        ...
11901  BMW           Z4  2016  premium unleaded (required)       240.0
11902  BMW           Z4  2016  premium unleaded (required)       335.0
11903  BMW           Z8  2001  premium unleaded (required)       394.0
11904  BMW           Z8  2002  premium unleaded (required)       394.0
11905  BMW           Z8  2003  premium unleaded (required)       394.0

       Engine Cylinders Transmission Type      Driven_Wheels  Number of Doors  \
0                   6.0            MANUAL   rear wheel drive              2.0
1                   6.0            MANUAL   rear wheel drive              2.0
2                   6.0            MANUAL   rear wheel drive              2.0
3                   6.0            MANUAL   rear wheel drive              2.0
4                   6.0            MANUAL   rear wheel drive              2.0
...                 ...               ...                ...              ...
11901               4.0            MANUAL   rear wheel drive              2.0
11902               6.0  AUTOMATED_MANUAL   rear wheel drive              2.0
11903               8.0            MANUAL   rear wheel drive              2.0
11904               8.0            MANUAL   rear wheel drive              2.0
11905               8.0            MANUAL   rear wheel drive              2.0

                           Market Category Vehicle Size Vehicle Style  \
0      Factory Tuner,Luxury,High-Performance       Compact         Coupe
```

```
1                          Luxury,Performance       Compact   Convertible
2                    Luxury,High-Performance        Compact         Coupe
3                          Luxury,Performance       Compact         Coupe
4                                     Luxury        Compact   Convertible
...                                      ...           ...           ...
11901                      Luxury,Performance       Compact   Convertible
11902                Luxury,High-Performance        Compact   Convertible
11903         Exotic,Luxury,High-Performance        Compact   Convertible
11904         Exotic,Luxury,High-Performance        Compact   Convertible
11905         Exotic,Luxury,High-Performance        Compact   Convertible

       highway MPG  city mpg  Popularity    MSRP
0               26         19        3916   46135
1               28         19        3916   40650
2               28         20        3916   36350
3               28         18        3916   29450
4               28         18        3916   34500
...            ...        ...         ...     ...
11901           34         22        3916   49700
11902           24         17        3916   66350
11903           19         12        3916  128000
11904           19         12        3916  130000
11905           19         12        3916  131500

[307 rows x 16 columns]
```

Output of: automatic_cars = df[

```
CARS WITH AUTOMATIC TRANSMISSION:
          Make    Model  Year                Engine Fuel Type   Engine HP  \
19        Audi      100  1992                regular unleaded       172.0
23        Audi      100  1993                regular unleaded       172.0
27        Audi      100  1994                regular unleaded       172.0
30        Audi      100  1994                regular unleaded       172.0
31        Audi      100  1994                regular unleaded       172.0
...        ...      ...   ...                             ...         ...
11909    Acura      ZDX  2012     premium unleaded (required)       300.0
11910    Acura      ZDX  2012     premium unleaded (required)       300.0
11911    Acura      ZDX  2012     premium unleaded (required)       300.0
11912    Acura      ZDX  2013  premium unleaded (recommended)       300.0
11913  Lincoln   Zephyr  2006                regular unleaded       221.0

       Engine Cylinders  Transmission Type       Driven_Wheels   Number of Doors  \
19                  6.0         AUTOMATIC    all wheel drive               4.0
23                  6.0         AUTOMATIC    all wheel drive               4.0
27                  6.0         AUTOMATIC   front wheel drive              4.0
30                  6.0         AUTOMATIC   front wheel drive              4.0
31                  6.0         AUTOMATIC    all wheel drive               4.0
...                 ...               ...                ...              ...
11909               6.0         AUTOMATIC    all wheel drive               4.0
11910               6.0         AUTOMATIC    all wheel drive               4.0
11911               6.0         AUTOMATIC    all wheel drive               4.0
11912               6.0         AUTOMATIC    all wheel drive               4.0
11913               6.0         AUTOMATIC   front wheel drive              4.0

                 Market Category  Vehicle Size  Vehicle Style  highway MPG  \
19                        Luxury      Midsize          Wagon           20
23                        Luxury      Midsize          Wagon           20
```

```
27                          Luxury      Midsize           Wagon           21
30                          Luxury      Midsize           Sedan           22
31                          Luxury      Midsize           Wagon           21
...                            ...          ...             ...          ...
11909  Crossover,Hatchback,Luxury      Midsize  4dr Hatchback           23
11910  Crossover,Hatchback,Luxury      Midsize  4dr Hatchback           23
11911  Crossover,Hatchback,Luxury      Midsize  4dr Hatchback           23
11912  Crossover,Hatchback,Luxury      Midsize  4dr Hatchback           23
11913                       Luxury      Midsize           Sedan           26

       city mpg  Popularity   MSRP
19           16        3105   2000
23           16        3105   2000
27           16        3105   2000
30           16        3105   2000
31           16        3105   2000
...         ...         ...    ...
11909        16         204  46120
11910        16         204  56670
11911        16         204  50620
11912        16         204  50920
11913        17          61  28995

[8266 rows x 16 columns]
```

Output of: selected_columns = df[['Make', 'MSRP']]

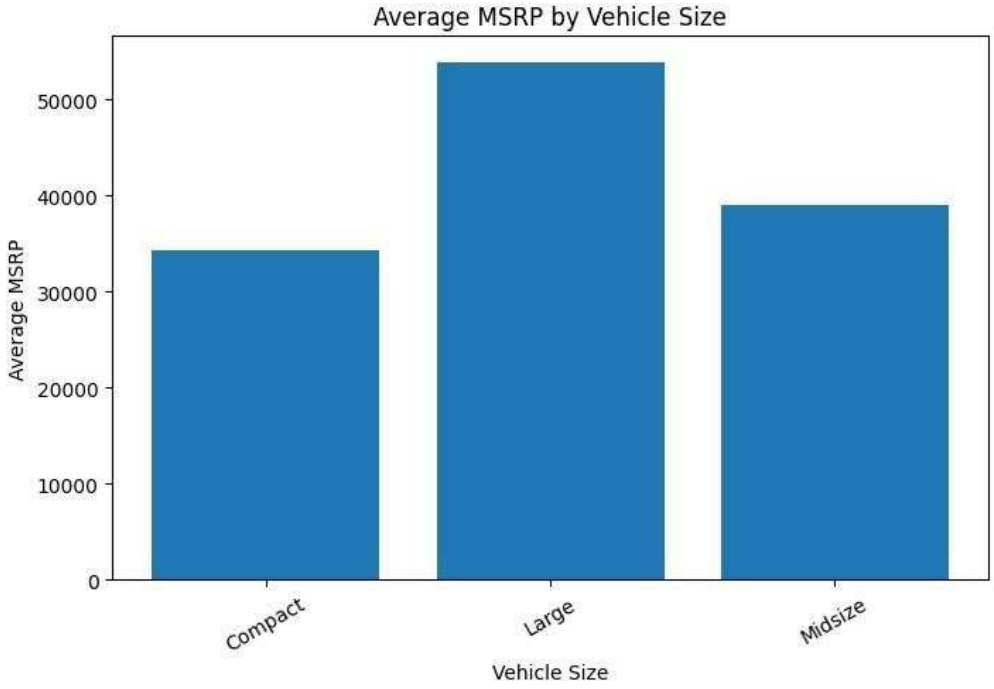
```
MAKE AND MSRP:
          Make   MSRP
0          BMW  46135
1          BMW  40650
2          BMW  36350
3          BMW  29450
4          BMW  34500
...        ...    ...
11909    Acura  46120
11910    Acura  56670
11911    Acura  50620
11912    Acura  50920
11913  Lincoln  28995

[11914 rows x 2 columns]
```

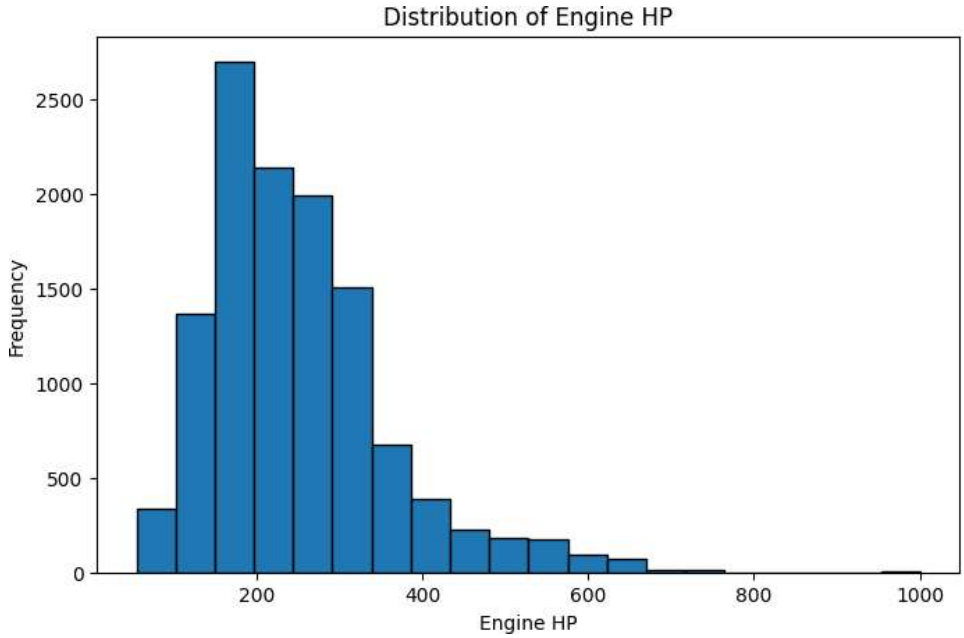
Output of: numeric_columns = [

```
DESCRIPTIVE STATISTICS OF SELECTED COLUMNS:
         Engine HP           MSRP       city mpg    highway MPG
count  11845.00000   1.191400e+04  11914.000000  11914.000000
mean     249.38607   4.059474e+04     19.733255     26.637485
std      109.19187   6.010910e+04      8.987798      8.863001
min       55.00000   2.000000e+03      7.000000     12.000000
25%      170.00000   2.100000e+04     16.000000     22.000000
50%      227.00000   2.999500e+04     18.000000     26.000000
75%      300.00000   4.223125e+04     22.000000     30.000000
max     1001.00000   2.065902e+06    137.000000    354.000000
```

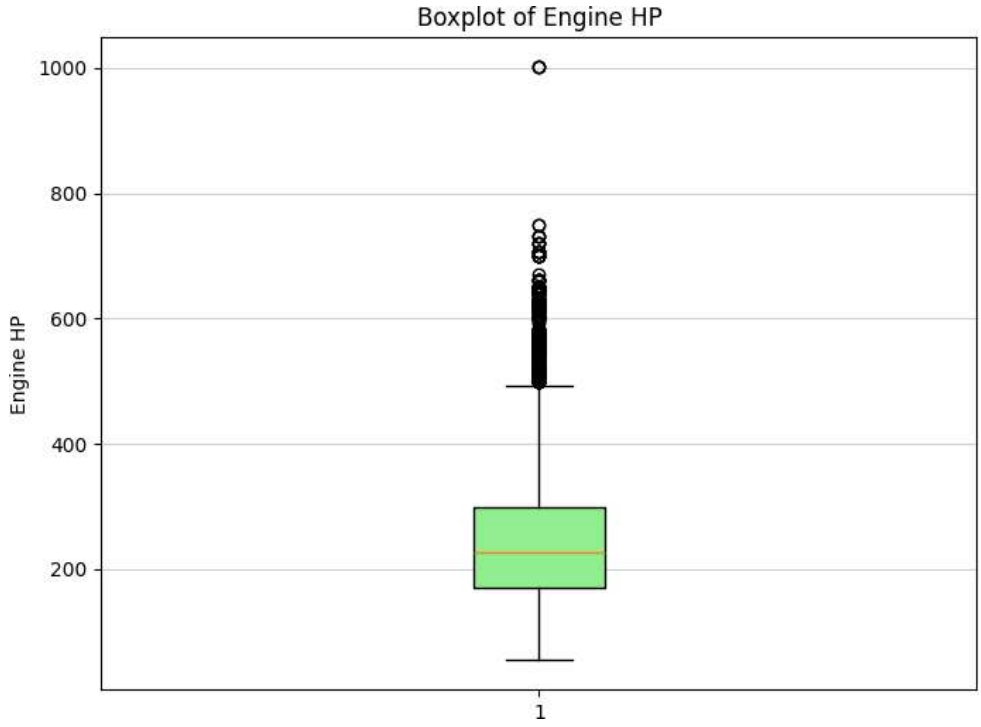
Output of: average_msrp = df.groupby('Vehicle Size')['MSRP'].mean()



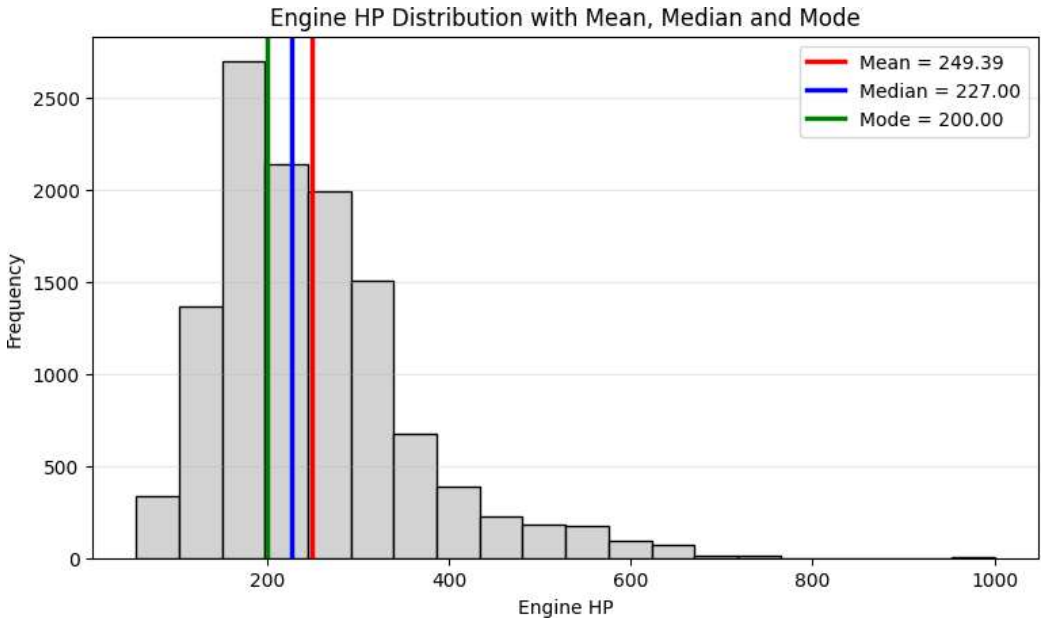
Output of: hp_data = df['Engine HP'].dropna()



Output of: plt.figure(figsize=(8,6))



Output of: mean_hp = hp_data.mean()



RESULT:

Thus, the car dataset (11914 rows and 16 columns) was inspected and analysed. The data was filtered using conditions, descriptive statistics were calculated (for example, the mean Engine HP is 249.39 and the median is 227), and the bar chart, histogram, box plot and mean-median-mode plot were drawn.